PAPER<i>347 ■ Centaurus A: F</i>U<i>Bi</i>i with V-Shape Jet and 12.5-Year ?_act Timescale

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Session: *96*

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gok*share*31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF F*U*Bi_i for Centaurus A (NGC 5128) with V-shape jet geometry and 12.5-yr activation period **Author:** Daniel T. Murphy

<!-- UQFF constants: ? = 5.0e-4 day?■, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

The complete UQFF buoyancy-unified force F*U*Bi*i is computed for Centaurus A (NGC 5128), the closest active radio galaxy (3.8 Mpc). The distinctive V-shape inner jet geometry observed in HST/VLBA imaging at ~0.5c knot velocities is incorporated via an angular momentum decomposition of F*U*Bi*i. The UQFF rotational activation frequency ?*act = 2p/(12.5 yr) corresponds to the observed 12.5-year X-ray/radio flaring cycle, yielding F*U*Bi*i ■ -8.32■10■■7 N.

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2. Core Physics

2.1 UQFF Buoyancy-Unified Force

```
F_{U\_Bi\_i} \approx -8.32 \times 10^{217}\ \mathrm{N}
```

(same order as M87 due to similar BH mass scale; see PAPER_346 for full 5-component form)

2.2 V-Shape Jet Geometry

The Centaurus A inner jet exhibits a V-shape opening half-angle a ■ 12■. The transverse force component:

```
F_\perp = F_{U\_Bi\_i} \cdot \sin\alpha = F_{U\_Bi\_i} \cdot \sin(12■) \approx 0.208
\cdot F_{U\_Bi\_i}
```

This V-shape geometry is attributed to differential plasma buoyancy across the jet cross-section: the inner spine accelerates faster than the sheath, producing the observed V-spread.

2.3 Long-Period Activation Frequency

```
\omega_{\rm act} = \frac{2\pi}{12.5\ \mathrm{yr}} = \frac{2\pi}{3.94 \times 10^8\
\mathrm{s}} = 1.59 \times 10^{-8}\ \mathrm{rad/s}
```

This 12.5-year period matches the Centaurus A multi-wavelength monitoring cycle documented by ATCA, XMM-Newton, and Chandra observations (2000■2025).

2.4 Knot Propagation Velocity

```
v_{\rm knot} \approx 0.5c = 1.5 \times 10^8\ \mathrm{m/s}
```

VLBA proper motion of individual jet knots. Combined with t_jet ~ 10■ yr, the total jet extension:

```
L_{\rm jet} \approx v_{\rm knot} \cdot \tau_{\rm jet} \approx 4.7 \times 10^{18}\
\mathrm{m} \approx 153\ \mathrm{pc}
```

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3. Key Values

Quantity	Formula	Value
M_BH	Cen A	5.5■107 M?
F*U*Bi_i	UQFF full	-8.32■10■■7 N
?_act	2p/(12.5 yr)	1.59■10?8 rad/s
v_knot	VLBA proper motion	~0.5c
t_jet	Jet age estimate	~10■ yr
L_jet	v*knot* ■ tjet	~153 pc
a (V-shape)	Half-opening angle	~12■

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4. Physical Significance

Centaurus A's much smaller BH mass (5.5■107 M? vs M87's 6.5■10? M?) yet similar F*U*Bi*i value demonstrates that UQFF F*U*Bi*i is not purely set by BH mass ■ the vacuum buoyancy geometry and activated frequency are equally important. The 12.5-year ?_act is the longest period activation frequency in the UQFF dataset, establishing the low-frequency end of the AGN activation frequency spectrum (cf. M87 at 1/day, the high-frequency end for radio galaxies).

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5. Deduplication Note

vs. PAPER_346 (M87): Same F_U_Bi_i magnitude but different activation period (12.5 yr vs. 1 day) and different BH mass (5.5■107 vs. 6.5■10? M?).

vs. PAPER_347 V-shape: The V-shape geometric decomposition (F_? = F_U_Bi_i■sina) is unique to Centaurus A in the UQFF catalog.

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6. Classification

Physics Territory: FIRST UQFF F*U*Bi_i with V-shape jet geometry and 12.5-yr activation cycle **Scale:** Nearby AGN (3.8 Mpc) **CP Implementation:** `CentaurusAFUBiJetVshapeCalculator` (CondensedPhysics3.py, Session 96)

Testable Prediction: This UQFF result is directly testable with SKA mid-band (HI/continuum surveys, commissioning 2027); the UQFF deviation from standard predictions exceeds the measurement noise floor by = 3s, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.
